# N4000-13EP - Dielectric Properties Table

**LAMINATE:**

| Thickness (inch) | & | Tol. | Construction | | | | | RC% | Nominal Thickness (inch) | Nominal Thickness (µm) | 1 GHz Dk | 2 GHz Dk | 2 GHz Df | 10 GHz Dk | 10 GHz Df |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0.0020** | ± | 0.0005 | 1 | 106 | | | | 72% | 0.0023 | 59 | 3.25 | 3.24 | 0.0085 | 3.20 | 0.0094 |
| **0.0025** | ± | 0.0005 | 1 | 1080 | | | | 57% | 0.0027 | 67 | 3.56 | 3.51 | 0.0081 | 3.49 | 0.0090 |
| **0.0030** | ± | 0.0005 | 1 | 1080 | | | | 61% | 0.0030 | 75 | 3.47 | 3.44 | 0.0082 | 3.41 | 0.0091 |
| **0.0030** | ± | 0.0005 | 1 | 2113 | | | | 44% | 0.0031 | 78 | 3.91 | 3.83 | 0.0076 | 3.82 | 0.0085 |
| **0.0035** | ± | 0.0005 | 1 | 2113 | | | | 49% | 0.0035 | 88 | 3.78 | 3.71 | 0.0078 | 3.69 | 0.0087 |
| **0.0035** | ± | 0.0005 | 1 | 106 | + | 1 | 106 | 65% | 0.0035 | 90 | 3.39 | 3.36 | 0.0083 | 3.33 | 0.0092 |
| **0.0040** | ± | 0.0005 | 1 | 106 | + | 1 | 106 | 68% | 0.0039 | 98 | 3.34 | 3.32 | 0.0084 | 3.28 | 0.0093 |
| **0.0040** | ± | 0.0005 | 1 | 1080 | + | 1 | 106 | 59% | 0.0042 | 107 | 3.52 | 3.49 | 0.0081 | 3.46 | 0.0090 |
| **0.0040** | ± | 0.0005 | 1 | 2116 | | | | 44% | 0.0041 | 105 | 3.90 | 3.82 | 0.0076 | 3.82 | 0.0085 |
| **0.0040** | ± | 0.0005 | 1 | 2113 | | | | 55% | 0.0041 | 105 | 3.61 | 3.56 | 0.0080 | 3.54 | 0.0089 |
| **0.0045** | ± | 0.0005 | 2 | 1080 | | | | 51% | 0.0044 | 113 | 3.71 | 3.65 | 0.0079 | 3.63 | 0.0088 |
| **0.0050** | ± | 0.0007 | 2 | 1080 | | | | 57% | 0.0053 | 135 | 3.56 | 3.51 | 0.0081 | 3.49 | 0.0090 |
| **0.0050** | ± | 0.0007 | 1 | 2113 | + | 1 | 106 | 55% | 0.0054 | 137 | 3.62 | 3.57 | 0.0080 | 3.54 | 0.0089 |
| **0.0060** | ± | 0.0007 | 2 | 1080 | | | | 61% | 0.0059 | 150 | 3.47 | 3.44 | 0.0082 | 3.41 | 0.0091 |
| **0.0060** | ± | 0.0007 | 2 | 2113 | | | | 44% | 0.0062 | 156 | 3.91 | 3.83 | 0.0076 | 3.82 | 0.0085 |
| **0.0070** | ± | 0.0010 | 2 | 2113 | | | | 49% | 0.0069 | 176 | 3.78 | 3.71 | 0.0078 | 3.69 | 0.0087 |
| **0.0080** | ± | 0.0010 | 2 | 2116 | | | | 44% | 0.0082 | 209 | 3.90 | 3.82 | 0.0076 | 3.82 | 0.0085 |
| **0.0100** | ± | 0.0010 | 2 | 2116 | | | | 54% | 0.0105 | 267 | 3.65 | 3.60 | 0.0080 | 3.58 | 0.0089 |
| **0.0120** | ± | 0.0015 | 2 | 1080 | + | 1 | 7628 | 46% | 0.0124 | 316 | 3.84 | 3.77 | 0.0077 | 3.76 | 0.0086 |

# N4000-13EP - Dielectric Properties Table

**LAMINATE (continued):**

| Thickness (inch) | & | Tol. | Construction | | | | | RC% | Nominal Thickness (inch) | Nominal Thickness (µm) | 1 GHz Dk | 2 GHz Dk | 2 GHz Df | 10 GHz Dk | 10 GHz Df |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0.0140** | ± | 0.0015 | 2 | 2113 | + | 1 | 7628 | 45% | 0.0146 | 372 | 3.87 | 3.80 | 0.0077 | 3.79 | 0.0085 |
| **0.0140** | ± | 0.0015 | 2 | 7628 | | | | 39% | 0.0143 | 362 | 4.07 | 3.97 | 0.0074 | 3.98 | 0.0083 |
| **0.0160** | ± | 0.0015 | 2 | 7628 | + | 1 | 1080 | 41% | 0.0165 | 418 | 4.02 | 3.93 | 0.0075 | 3.93 | 0.0083 |
| **0.0160** | ± | 0.0015 | 2 | 2116 | + | 1 | 7628 | 43% | 0.0159 | 405 | 3.93 | 3.85 | 0.0076 | 3.85 | 0.0085 |
| **0.0180** | ± | 0.0015 | 2 | 7628 | + | 1 | 2113 | 42% | 0.0184 | 467 | 3.96 | 3.88 | 0.0076 | 3.87 | 0.0084 |
| **0.0200** | ± | 0.0025 | 3 | 7628 | | | | 39% | 0.0214 | 543 | 4.07 | 3.97 | 0.0074 | 3.98 | 0.0083 |
| **0.0240** | ± | 0.0025 | 2 | 2116 | + | 2 | 7628 | 45% | 0.0248 | 629 | 3.89 | 3.81 | 0.0077 | 3.80 | 0.0085 |
| **0.0280** | ± | 0.0025 | 4 | 7628 | | | | 39% | 0.0285 | 724 | 4.07 | 3.97 | 0.0074 | 3.98 | 0.0083 |

# N4000-13EP - Dielectric Properties Table

**PREPREG:**

| Glass | RC% | Thickness (inches) | Thickness (µm) | 1 GHz Dk | 2 GHz Dk | 2 GHz Df | 10 GHz Dk | 10 GHz Df |
|---|---|---|---|---|---|---|---|---|
| **106** | 68.4% | 0.0020 | 51 | 3.32 | 3.30 | 0.0084 | 3.26 | 0.0093 |
| **106** | 72.2% | 0.0023 | 59 | 3.25 | 3.24 | 0.0085 | 3.20 | 0.0094 |
| **106** | 75% | 0.0026 | 66 | 3.20 | 3.19 | 0.0085 | 3.15 | 0.0095 |
| **1080** | 65% | 0.0034 | 86 | 3.39 | 3.37 | 0.0083 | 3.33 | 0.0092 |
| **1080** | 70% | 0.0041 | 103 | 3.29 | 3.28 | 0.0084 | 3.24 | 0.0094 |
| **2113** | 57% | 0.0044 | 111 | 3.57 | 3.52 | 0.0081 | 3.50 | 0.0090 |
| **2116** | 54% | 0.0053 | 135 | 3.64 | 3.59 | 0.0080 | 3.56 | 0.0089 |
| **7628** | 44% | 0.0080 | 204 | 3.91 | 3.83 | 0.0076 | 3.82 | 0.0085 |

These laminate and prepreg items are representative of available materials. These items and their associated values may not be regularly available and additional lead times or minimum order quantities could apply. Please contact our office or your local technical representative for assistance with choosing the best material for your design and application.

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